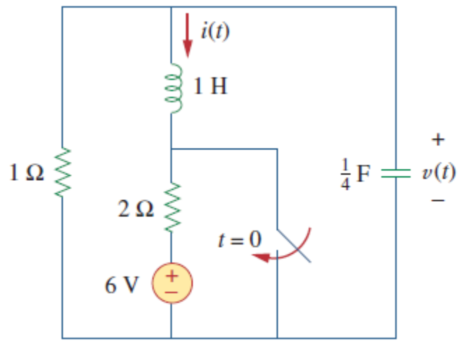
Answer all questions with detail steps.

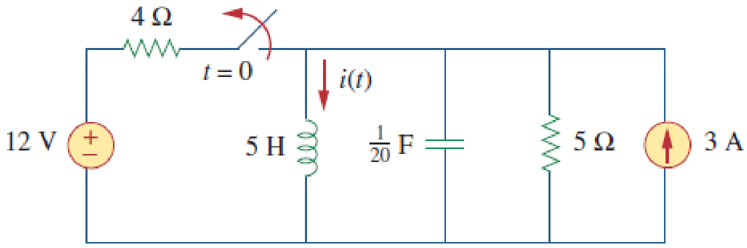
Q1/

Given the circuit in , find $i(t)$ and $v(t)$ for $t > 0$.



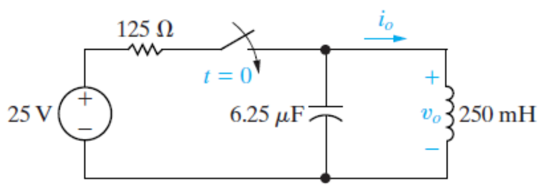
Q2/

Determine $i(t)$ for $t > 0$ in the circuit



Q3/

There is no energy stored in the circuit when the switch is closed at $t = 0$. Find $i_o(t)$ for $t \geq 0$.



Q4/

Design a series *RLC* circuit using component values from tables below, with a resonant

radian frequency of 20,000 rad/s .

1- Choose a resistor or create a resistor network so that the response is critically damped. Draw your circuit. Calculate the roots of the characteristic equation.
2- Choose a resistor or create a resistor network so that the response is underdamped. Draw your circuit. Calculate the roots of the characteristic equation.
3- Choose a resistor or create a resistor network so that the response is overdamped. Draw your circuit. Calculate the roots of the characteristic equation.

Resistors (5% tolerance) [Ω]					
10	100	1.0 k	10k	100k	1.0 M
	120	1.2 k	12 k	120 k	
15	150	1.5 k	15 k	150 k	1.5 M
	180	1.8 k	18 k	180 k	
22	220	2.2 k	22 k	220 k	2.2 M
	270	2.7 k	27 k	270 k	
33	330	3.3 k	33 k	330 k	3.3 M
	390	3.9 k	39 k	390 k	
47	470	4.7 k	47 k	470 k	4.7 M
	560	5.6 k	56 k	560 k	
68	680	6.8 k	68 k	680 k	6.8 M

Capacitors		
10 pF	22 pF	47 pF
100 pF	220 pF	470 pF
0.001 μF	0.0022 μF	0.0047 μF
0.01 μF	0.022 μF	0.047 μF
0.1 μF	0.22 μF	0.47 μF
1 μF	2.2 μF	4.7 μF
10 μF	22 μF	47 μF
100 μF	220 μF	470 μF